$m = \tan\theta$.

When θ is $\frac{\pi}{2}$, $\Rightarrow m = \tan\frac{\pi}{2}$ is undefined.

(ii) When (x_1, y_1) and (x_2, y_2) are any two points on the line with $x_2 \neq x_1$, then the slope is the change in the y coordinate divided by the corresponding change in the x coordinate.
This is described by the following equation.

$$m = \frac{\Delta y}{\Delta x} = \frac{y_2 - y_1}{x_2 - x_1} = \frac{vertical\ change}{horizontal\ change}$$

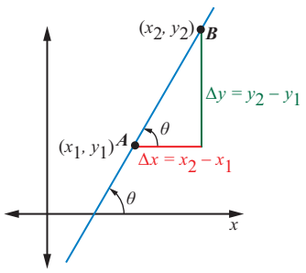


Figure 6.14

(iii) When the general form of the linear equation $ax + by + c = 0$ is given, then the slope of the line is

$$m = -\frac{a}{b}, \quad b \neq 0.$$

m is undefined when $b = 0$

The slope of a line can be a positive or negative or zero or undefined as shown below:

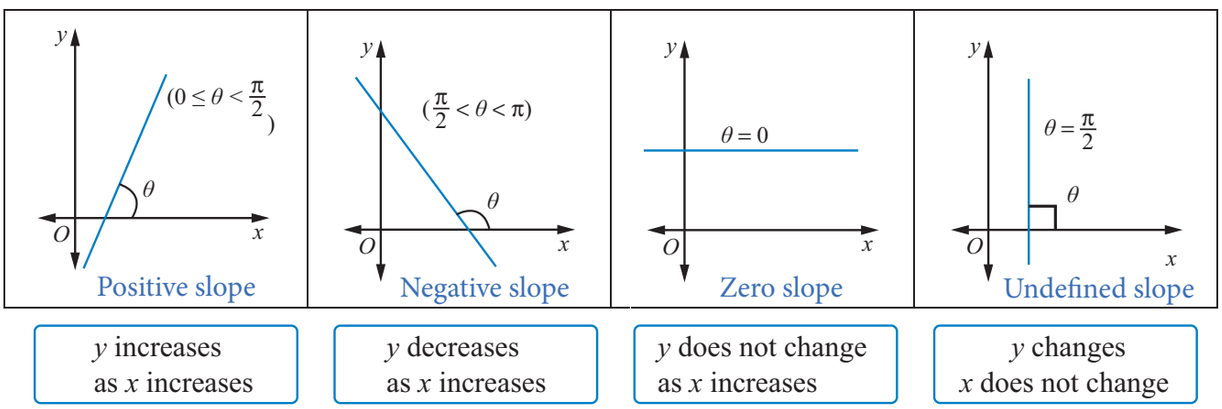


Figure 6.15

Definition 6.5

In a plane three or more points are said to be collinear if they lie on a same straight line.

Let A, $B$and C be any three points on a plane. If the slope of AB is equal to the slope of BC (or AC), then they are collinear.